Huawei TechArena 2025: Challenge Summary and Technical Plan

Coding Assistant Brief

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1 Challenge Context

Primary objective: Develop a BESS to optimize financial performance by participating in day-ahead, FCR and aFRR markets.

The challenge is divided into three core, interconnected optimization tasks:

1. **Operation Optimization:** Maximize the BESS's revenue over the year 2024 by developing an optimal charge/discharge strategy to bid in the DA, FCR and aFRR markets.
2. **Investment Optimization:** Identify which of five European countries (Germany, Austria, Switzerland, Hungary, Czech Republic) offers the highest Return on Investment (ROI) for installing the BESS over a 10-year period.
3. **Configuration Optimization:** Determine the optimal BESS configuration by analyzing the impact of different C-rates and daily cycle limits on profitability and performance.

Table 1: European Market Participation Rules Summary

Feature	Day-Ahead Market (EPEX SPOT)	FCR Capacity Market	aFRR Capacity Market
Mechanism	Blind Auction	Daily Auction	Daily Auction
Gate Closure Time (D-1)	12:00 CET	8:00 CET	9:00 CET
Product Granularity	15 minutes	4 hours	4 hours
Bid Structure	Energy (MWh)	Symmetric Capacity (MW)	Asymmetric Capacity (MW)
Remuneration	Pay-as-Cleared (Energy)	Pay-as-Cleared (Capacity)	Pay-as-Bid (Capacity)
Minimum Bid Size	0.1 MW	1 MW	1 MW

2 Evaluation Criteria

Submissions will be evaluated based on the following criteria:

1. Revenue maximization (50%). This will evaluate how well the algorithm maximizes revenue based on market prices. A linear scaling formula will be used to map all submission values between the minimum and maximum observed
2. Investment optimization (20%). This will evaluate the assessment of optimal investment locations and markets. The evaluation will consider
3. Configuration optimization (20%). This will evaluate the assessment of the most relevant configuration parameters on BESS revenue.
4. Code Quality and Documentation (10%). This will evaluate the clarity and structure of the code.

Table 2: BESS Technical and Configuration Parameters

Parameter	Symbol	Value	Unit	Source
Fixed Parameters				
Nominal Energy Capacity	E_{nom}	4472	kWh	
Rated Power	P_{rated}	2236	kW	
Initial Investment Cost	C_{inv}	200	EUR/kWh	
Configuration Scenarios				
C-rate = 0.25 C				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.25\text{C}}$	$0.25 \times 4472 = 1118$	kW	
C-rate = 0.33 C				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.33\text{C}}$	$0.33 \times 4472 \approx 1475.8$	kW	
C-rate = 0.50 C				
Max Power ($P_{\text{max}}^{\text{config}}$)	$P_{\text{max}}^{0.50\text{C}}$	$0.50 \times 4472 = 2236$	kW	
Cycles = 1.0/day				
Max Daily Discharge	$E_{\text{dis,max}}^{1.0}$	$1.0 \times 4472 = 4472$	kWh	
Cycles = 1.5/day				
Max Daily Discharge	$E_{\text{dis,max}}^{1.5}$	$1.5 \times 4472 = 6708$	kWh	
Cycles = 2.0/day				
Max Daily Discharge	$E_{\text{dis,max}}^{2.0}$	$2.0 \times 4472 = 8944$	kWh	

3 Project Plan

Outlook

The Interdependence of the Three Optimization Tasks A deeper analysis reveals a nested, hierarchical relationship between them that must be reflected in the project's modeling strategy.

The "Optimal Configuration" task is at the base of the hierarchy. The choice of a C-rate (0.5C, 0.33C, 0.25C) and a daily cycle limit (1.0, 1.5, 2.0) directly defines the physical constraints of the BESS, such as its maximum charge/discharge power and daily energy throughput. These parameters are fundamental inputs for the "Optimal Operation" model.

The **"Optimal Operation"** model forms the core of the analysis. It takes the configuration parameters as given and, based on market price data, calculates the maximum possible annual revenue for that specific BESS configuration in a given country. This revenue figure is the most critical output of the entire project.

Finally, the "Optimal Investment" task sits at the top of the hierarchy. It uses the annual revenue generated by the operational model as its primary input to calculate the 10-year ROI. The optimal country for investment is simply the one that yields the highest ROI.

The Nested Hierarchy of the Three Optimization Tasks

1. **"Optimal Operation"**: Focus on building a robust and accurate operational optimization model. This model will then be placed within a larger computational framework that systematically iterates through every possible BESS configuration and every target country.

2. **"Optimal Configuration"**: By running the operational model for each combination (e.g., Germany with 0.25C and 1.0 cycle, Germany with 0.5C and 1.5 cycles, Austria with 0.25C and 1.0 cycle, etc.), the model can then populate a complete results matrix.

3. **"Optimal Investment"**: Use the annual revenue generated by the operational model as its primary input to calculate the 10-year ROI. The optimal country for investment is simply the one that yields the highest ROI considering WACC and inflation rates.

Mathematical Model

Objective Function

The objective is to maximize the total net profit over the one-year horizon. This is the sum of day-ahead energy arbitrage revenue and ancillary service capacity payments, minus the cost of energy purchased for charging.

$$\begin{aligned}\max\ Z = &\sum_{t\in T}\left(\frac{P_{DA}(t)}{1000}\,p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000}\,p_{\text{ch}}(t)\right)\Delta t \\ &+ \sum_{b\in B}\Big(P_{FCR}(b)\,c_{fcr}(b) + P^{\text{pos}}_{aFRR}(b)\,c^{\text{pos}}_{aFRR}(b) + P^{\text{neg}}_{aFRR}(b)\,c^{\text{neg}}_{aFRR}(b)\Big)\,\Delta b \end{aligned} \tag{1}$$

- The first term is day-ahead net profit. $p_{\text{dis}}(t)$ and $p_{\text{ch}}(t)$ are in kW, $P_{DA}(t)$ is in EUR/MWh; division by 1000 converts kW to MW. Multiplying by Δt (0.25 h) yields EUR per interval.
- The second term is ancillary service capacity revenue. Bids $c(b)$ are in MW and prices $P(b)$ are in EUR/MW for each 4-hour block; multiplying by Δb (4 h) yields EUR per block.

Constraint Equations with Detailed Explanations

The optimization is subject to the following constraints.

1. **State of Charge (SOC) Dynamics:** Energy balance of the BESS.

$$e_{\text{soc}}(t) = e_{\text{soc}}(t-1) + \left(p_{\text{ch}}(t)\,\eta_{\text{ch}} - \frac{p_{\text{dis}}(t)}{\eta_{\text{dis}}}\right)\Delta t \qquad \forall t\in T \tag{2}$$

For $t = 1$, use the initial SOC:

$$e_{\text{soc}}(1) = e^{\text{init}}_{\text{soc}} + \left(p_{\text{ch}}(1)\,\eta_{\text{ch}} - \frac{p_{\text{dis}}(1)}{\eta_{\text{dis}}}\right)\Delta t. \tag{3}$$

2. **SOC Limits:** Stay within operational boundaries.

$$SOC_{\min}\,E_{\text{nom}} \le e_{\text{soc}}(t) \le SOC_{\max}\,E_{\text{nom}} \qquad \forall t\in T \tag{4}$$

3. **Power Limits:** Binary–continuous linking to configured power.

$$p_{\text{ch}}(t) \le y_{\text{ch}}(t)\,P^{\text{config}}_{\max} \qquad \forall t\in T \tag{5}$$

$$p_{\text{dis}}(t) \le y_{\text{dis}}(t)\,P^{\text{config}}_{\max} \qquad \forall t\in T \tag{6}$$

4. **Simultaneous Operation Prevention:** No charge and discharge at the same time.

$$y_{\text{ch}}(t) + y_{\text{dis}}(t) \le 1 \qquad \forall t\in T \tag{7}$$

5. **Market Co-optimization Power Limits:** Allocate total power between energy and reserves (convert MW bids to kW by $\times 1000$).

$$p_{\text{ch}}(t) + 1000\,c_{fcr}(b) + 1000\,c^{\text{pos}}_{aFRR}(b) \le P^{\text{config}}_{\max} \qquad \forall b\in B,\ \forall t\in b \tag{8}$$

$$p_{\text{dis}}(t) + 1000\,c_{fcr}(b) + 1000\,c^{\text{neg}}_{aFRR}(b) \le P^{\text{config}}_{\max} \qquad \forall b\in B,\ \forall t\in b \tag{9}$$

6. **Daily Cycle Limit:** Limit the total daily discharged energy.

$$\sum_{t\in d} p_{\text{dis}}(t)\,\Delta t \ \le\ N_{\text{cycles}}\,E_{\text{nom}} \qquad \forall d\in D \tag{10}$$

7. **Ancillary Service Energy Reserve:** Maintain sufficient energy to deliver awarded capacity for 15 minutes ($\Delta t = 0.25$ h).

$$e_{\text{soc}}(t) \geq SOC_{\min} E_{\text{nom}} + \left(1000\, c_{fcr}(b) + 1000\, c^{\text{pos}}_{aFRR}(b)\right) \Delta t \qquad \forall b \in B,\, \forall t \in b \tag{11}$$

$$e_{\text{soc}}(t) \leq SOC_{\max} E_{\text{nom}} - \left(1000\, c_{fcr}(b) + 1000\, c^{\text{neg}}_{aFRR}(b)\right) \Delta t \qquad \forall b \in B,\, \forall t \in b \tag{12}$$

8. **Minimum Bid Size Constraints:** Bids can be non-zero only if the corresponding binary is 1; if non-zero, they must satisfy minimum size and available power.

$$c_{fcr}(b) \geq y_{fcr}(b)\, \text{MinBid}_{fcr} \qquad \forall b \in B \tag{13}$$

$$c_{fcr}(b) \leq y_{fcr}(b)\, \frac{P^{\text{config}}_{\max}}{1000} \qquad \forall b \in B \tag{14}$$

$$c^{\text{pos}}_{aFRR}(b) \geq y^{\text{pos}}_{aFRR}(b)\, \text{MinBid}_{afrr} \qquad \forall b \in B \tag{15}$$

$$c^{\text{pos}}_{aFRR}(b) \leq y^{\text{pos}}_{aFRR}(b)\, \frac{P^{\text{config}}_{\max}}{1000} \qquad \forall b \in B \tag{16}$$

$$c^{\text{neg}}_{aFRR}(b) \geq y^{\text{neg}}_{aFRR}(b)\, \text{MinBid}_{afrr} \qquad \forall b \in B \tag{17}$$

$$c^{\text{neg}}_{aFRR}(b) \leq y^{\text{neg}}_{aFRR}(b)\, \frac{P^{\text{config}}_{\max}}{1000} \qquad \forall b \in B \tag{18}$$

9. **Ancillary Service Market Mutual Exclusivity:** Prevent simultaneous bidding in multiple ancillary service markets for the same block.

$$y_{fcr}(b) + y^{\text{pos}}_{aFRR}(b) + y^{\text{neg}}_{aFRR}(b) \leq 1 \qquad \forall b \in B \tag{19}$$

4 Investment Optimization

This section outlines the investment optimization framework used to evaluate the financial viability of deploying a Battery Energy Storage

4.1 Objective

The primary objective of the investment optimization is to evaluate and rank the financial viability of deploying a Battery Energy Storage System (BESS) across various European countries and technical configurations. This is achieved by performing a 10-year Discounted Cash Flow (DCF) analysis to determine the Net Present Value (NPV) and the Levelized Return on Investment (ROI) for each scenario.

4.2 Financial Modeling Approach

As established, the core of our analysis is a DCF model that correctly aligns nominal cash flows with a nominal discount rate. This approach ensures that the effects of inflation are accounted for consistently and accurately.

- **Cash Flows:** We project future profits by growing the initial year's profit (from the operational optimization) at the country-specific inflation rate. This creates a stream of **nominal cash flows**.
- **Discount Rate:** We use the country-specific Weighted Average Cost of Capital (WACC) as the **nominal discount rate**.

4.3 Step-by-Step Calculation

The analysis follows a three-step process for the results from 45 scenarios (5 countries × 9 configurations).

4.3.1 Step 1: Projecting Nominal Profits

First, for each country among the five, we take the BEST annual profit of this country (that is, the highest annual profit of nine different configuration settings) calculated by the Pyomo model for the year 2024, denoted as Π_{2024}, and project it over a 10-year horizon from year $y = 1$ (2024) to $y = 10$ (2033). The nominal profit for any future year y is calculated by applying the country's annual inflation rate, π.

$$\Pi_y = \Pi_{2024} \cdot (1 + \pi)^{y-1} \tag{20}$$

4.3.2 Step 2: Calculating Net Present Value (NPV)

Next, we calculate the NPV by summing the discounted values of all future nominal profits and subtracting the initial Capital Expenditure (CAPEX). The CAPEX is given as \$200 per kWh. For a 1 MW BESS with a nominal energy capacity E_{nom} (in MWh), the CAPEX is $200 \times E_{nom} \times 1000$.

The WACC is denoted by i.

$$\text{NPV} = \left(\sum_{y=1}^{10} \frac{\Pi_y}{(1+i)^y} \right) - \text{CAPEX} \tag{21}$$

4.3.3 Step 3: Calculating Levelized Return on Investment (ROI)

Finally, we calculate the Levelized ROI as specified in the competition guidelines. This metric annualizes the return relative to the initial investment.

$$\text{Levelized ROI}(\%) = \frac{\text{PV(Total Profits)}}{\text{CAPEX} \times \text{Lifetime}} \times 100 \tag{22}$$

Where:

- PV(Total Profits) is the first term in the NPV equation: $\sum_{y=1}^{10} \frac{\Pi_y}{(1+i)^y}$.
- CAPEX is the initial investment cost.
- Lifetime is the project duration, which is 10 years.

4.4 Example Calculation

Let's consider a hypothetical scenario for a 1 MW BESS:

- **BESS Configuration:** C-rate = 0.5 C $\implies E_{nom} = 2$ MWh
- **Initial CAPEX:** $200\,\text{EUR/kWh} \times 2000\,\text{kWh} = 400{,}000\,\text{EUR}$
- **Country Parameters:** WACC (i) = 7.0%, Inflation (π) = 2.0%
- **Simulated Profit (2024):** $\Pi_{2024} = 50{,}000\,\text{EUR}$

The Levelized ROI is then:

$$\text{Levelized ROI} = \frac{395{,}437.12}{400{,}000 \times 10} \times 100 = \frac{395{,}437.12}{4{,}000{,}000} \times 100 = 9.89\%$$

This result indicates that, for this specific scenario, the project yields an average annual return of 9.89% on its initial investment in present value terms, but it does not break even (NPV < 0).

5 Appendix: Knowledge Hub

5.1 The Asset: Mastering the the Huawei LUNA2000-4.5MWH-2H1 Smart String ESS

Fixed parameters:

- Nominal Capacity: 4,472 kWh
- Rated Power: 2,236 kW

5.2 Terminorlogy

- Investment Terminology:
  - ROI (Return on Investment):
  - WACC (Weighted Average Cost of Capital):
  - Inflation Rate:
- Market Terminology:
  - Day-Ahead Wholesale Market: A blind auction that closes at 12:00 CET on the day before delivery (Day D-1). It determines the price for all 15-minute intervals of the following day (Day D).
  - Frequency Containment Reserve (FCR): A daily auction for capacity, not energy. The auction closes at 8 a.m. on Day $D-1$ for six distinct 4-hour product blocks covering day D. Bids must be symmetric, meaning the BESS must offer an euqal amount of capacity for both positive (discharging to counter low frequency) and negative (charging to counter high frequency) reserves. The minimum bid size is 1 MW.
  - Automatic Frequency Restoration Reserve (aFRR): The aFRR market is a two-stage process involving a capacity market and an energy market. The Balancing Capacity Market (BCM) auction closes at 9:00 on Day $D-1$ for 4-hour product blocks. The Balancing Energy Market (BEM) involves near-real-time auctions that close 25 minutes before each 15-minute delivery interval. Unlike FCR, aFRR bids are asymmetric, allowing the BESS to offer different amounts of capacity for positive (upward) and negative (downward) reserves. The minimum bid size is 1 MW.

5.3 What is BESS and EMS?

5.3.1 Technical Overview

At its core, a Battery Energy Storage System (BESS) is a large, rechargeable battery. Its main job on the power grid is to absorb electrical energy, store it, and then release it when it's most needed or most profitable.

If the BESS is the muscle, the Energy Management System (EMS) is the brain that tells the BESS what to do and when. It constantly analyzes data, such as:

- Market Electricity Prices
- Battery's State of Charge (SoC)
- Grid Frequency
- Physical Limits of the Battery (like C-rate)

A BESS **cannot simultaneously charge and discharge**. At any given moment, the net flow of power is either into the battery (charging) or out of the battery (discharging). While it can switch between these states very quickly (often in milliseconds), they are mutually exclusive actions.

Besides just buying and selling in the day-ahead market, your BESS can make money in another, more subtle way: by helping to keep the grid stable. These are called **ancillary services**, and they include things like Frequency Containment Reserve (FCR) and automatic Frequency Restoration Reserve (aFRR):

1. **Frequency Containment Reserve (FCR):** The grid has a "heartbeat" or frequency (50 Hz in Europe). FCR is the ultra-fast, automatic response to tiny deviations in this frequency. A BESS providing FCR is paid to be on standby, ready to inject or absorb tiny amounts of power instantly to stabilize the grid's heartbeat.
2. **Automatic Frequency Restoration Reserve (aFRR):** This is the second line of defense. If a bigger event happens (like a power plant suddenly going offline) and the frequency shifts more significantly, aFRR is activated to push it back to the target 50 Hz.

*FCR is like the **cruise control** in a car. It makes constant, tiny, automatic adjustments to the engine to keep your speed perfectly steady, without you having to think about it; aFRR is you, the driver, noticing you're slowing down on a hill and gently pressing the accelerator to get back to your desired speed. It's a slightly slower but more powerful and deliberate correction.*

These two parameters define the physical capabilities and warranty constraints of the BESS

1. **C-Rate:** The C-rate is a measure of how quickly a battery can be charged or discharged relative to its total capacity. A 1C rate means the battery can be fully charged or discharged in one hour. A 0.5C rate means it would take two hours, etc.
2. **(Daily) Cycle Limit:** This is a warranty constraint that limits how much you can use the battery each day to prevent it from degrading too quickly. One "cycle" is equivalent to one full charge and one full discharge. Your project mentions limits of 1.0, 1.5, and 2.0. A limit of 1.0 means you can use the battery's full capacity once per day. A limit of 2.0 means you could, for instance, fully charge and discharge it twice.

5.3.2 Economic Overview - ROI, WACC, & Inflation

These three metrics help you compare the project's long-term profitability across the five different European countries.

1. **Return on Investment (ROI):** This is the ultimate scorecard for your project. It's a simple ratio that tells you how much profit you made relative to the initial cost. *Formula:*

$$\text{ROI} = \frac{\text{Total Profit over 10 years}}{\text{Initial Investment Cost}} \times 100\%$$

   Your goal is to find the country that offers the highest ROI over a 10-year period.
2. **Weighted-Average Cost of Capital (WACC):** This is the average interest rate a company has to pay to borrow the money needed to build the BESS. *Analogy:* Think of it as the project's "cost of money." A country with a high WACC is a more expensive place to finance the project. A higher WACC will eat into your profits and therefore lower your ROI.
3. **Inflation Rate:** This is the rate at which money loses its value over time. EUR 100 in your pocket today will buy less in a year. When you calculate your 10-year revenue, you have to account for inflation. The profits you make in Year 9 are worth less than the profits you make in Year 1. A country with high inflation will see its future revenues become less valuable more quickly.

Table 3: Model Sets, Parameters, and Decision Variables

Symbol	Definition	Unit	Type
Sets & Indices			
T	Set of 15-minute time intervals, $t \in T = \{1, ..., 35040\}$	-	Set
B	Set of 4-hour ancillary service blocks, $b \in B$	-	Set
D	Set of 24-hour days, $d \in D$	-	Set
Parameters			
$P_{DA}(t)$	Day-ahead electricity price in interval t	EUR/MWh	Input
$P_{FCR}(b)$	FCR capacity price in block b	EUR/MW	Input
$P_{aFRR}^{pos}(b)$	Positive aFRR capacity price in block b	EUR/MW	Input
$P_{aFRR}^{neg}(b)$	Negative aFRR capacity price in block b	EUR/MW	Input
E_{nom}	Nominal energy capacity of the BESS	kWh	Input
P_{max}^{config}	Maximum charge/discharge power for the selected configuration	kW	Input
η_{ch}, η_{dis}	Charging and discharging efficiencies	-	Input
SOC_{min}, SOC_{max}	Min/max state of charge as a fraction of E_{nom}	-	Input
N_{cycles}	Daily cycle limit for the selected configuration	-	Input
e_{nom}^{init}	Initial state of charge at the start of the simulation period	kWh	Input
Δt	Duration of a time interval (0.25)	h	Constant
Δb	Duration of an ancillary service block (4)	h	Constant
MinBid_{fcr}	Minimum bid size for FCR market	MW	Input
MinBid_{afrr}	Minimum bid size for aFRR market	MW	Input
Decision Variables			
$p_{ch}(t)$	Power used to charge the BESS in interval t	kW	Continuous ≥ 0
$p_{dis}(t)$	Power discharged from the BESS in interval t	kW	Continuous ≥ 0
$e_{soc}(t)$	Energy stored in the BESS at the	kWh	Continuous ≥ 0

WACC	Value		
Inflation Rate	value		
Discount Rate	Value		
Yearly Profits (2024)	Value		
Year		Initial Investment [kEUR/MWh]	Yearly profits [kEUR/MWh]
	2023		
	2024		
	2025		
	2026		
	2027		
	2028		
	2029		
	2030		
	2031		
	2032		
	2033		